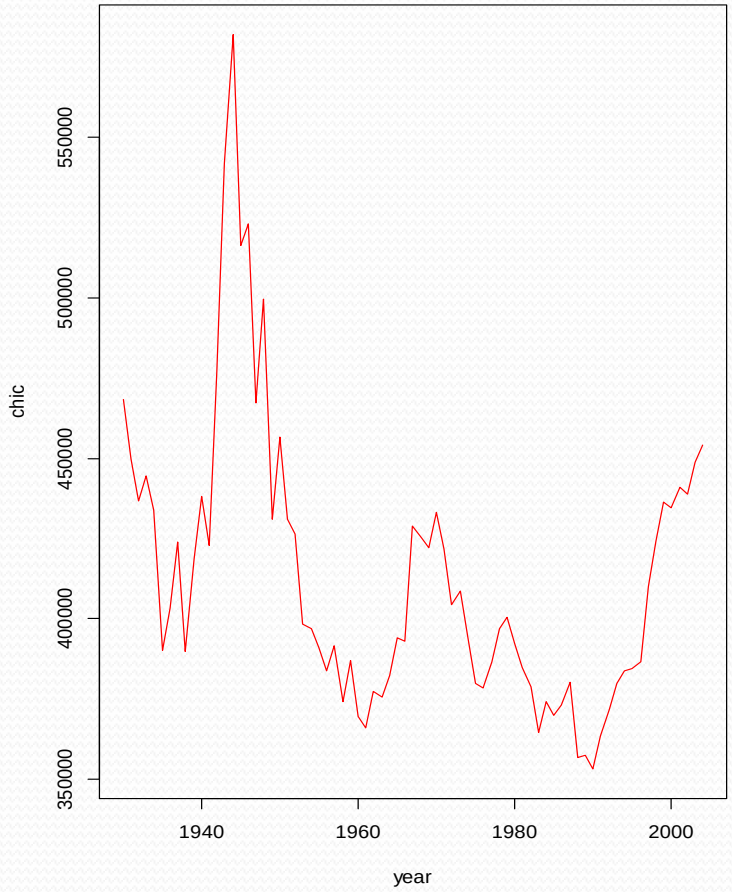
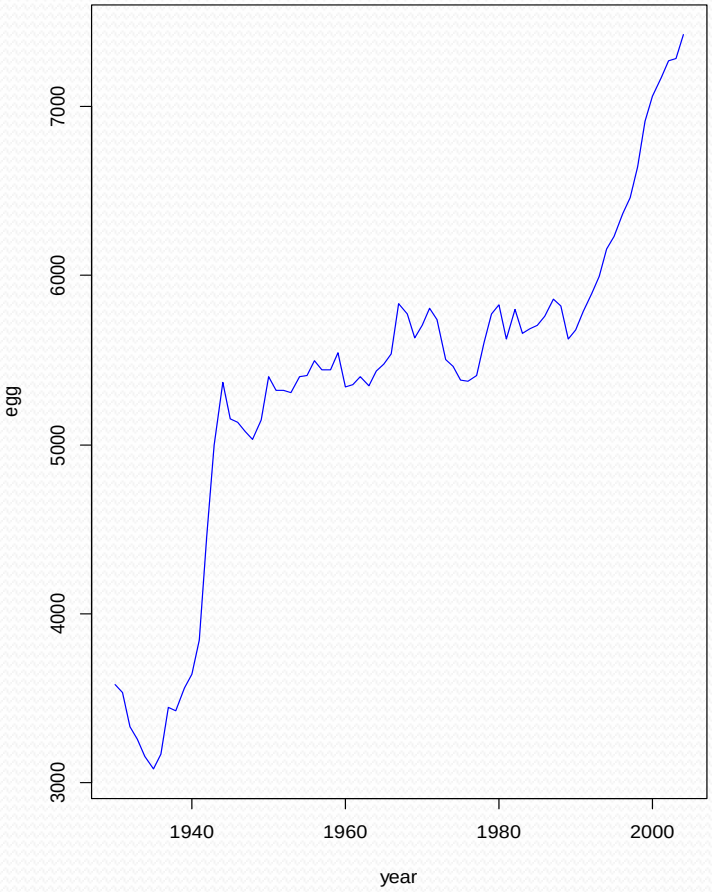
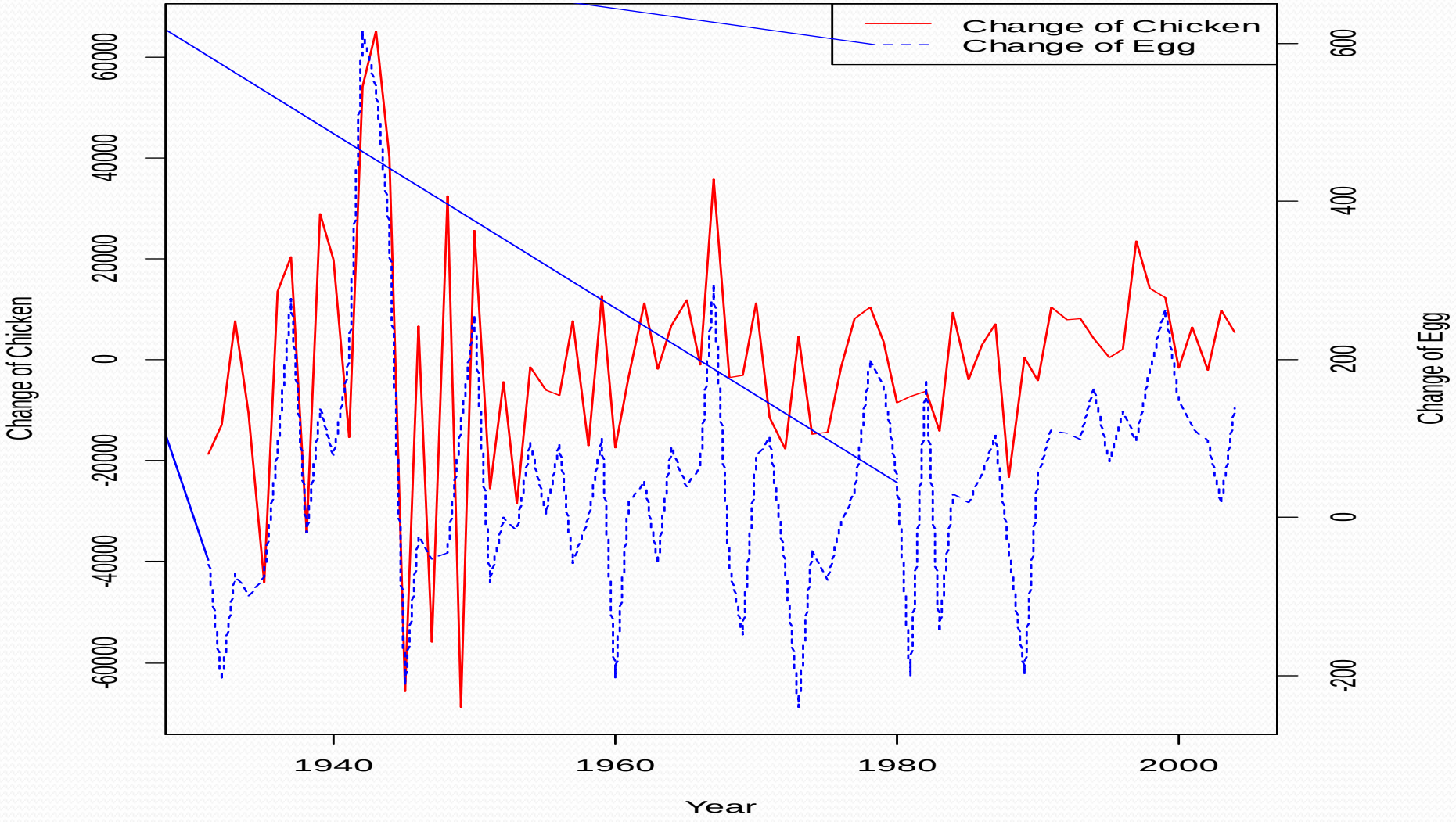
Q1-PS3 Fanglin Ye

Data Description





Data Description



Chicken or Eggs: Which come first?

- We would say that Y_t does not Granger cause X_t iff

$$E(X_t|\Omega) = E(X_t|\mathrm{X}_{\mathrm{t}-1}, X_{t-2}, \dots \dots)$$

$$where\ \Omega_t = \{\mathrm{X}_{\mathrm{t}-1}, X_{t-2}, \dots \dots Y_{t-1}, Y_{t-2} \dots \dots.\}$$

$$Egg_t = \alpha + \sum_{i=1}^{L} \beta_i Egg_{t-i} + \sum_{i=1}^{L} \gamma_i Chicken_{t-i}$$

$$Chicken_t = \alpha + \sum_{i=1}^{L} \beta_i Chicken_{t-i} + \sum_{i=1}^{L} \gamma_i Egg_{t-i}$$

$$H_0: \gamma_1 = \gamma_2 = \cdots = 0$$

Test Results

	Chicken for egg		Egg for Chicken	
L	F	P-value	F	P-value
1	0.0073	0.9321	0.0753	0.785
2	0.6414	0.5296	9.982514	0.000153***
3	0.4657	0.7072	5.596	0.00174***
4	0.3191	0.8641	4.5476	0.003***

Conclusion

- it is obvious to see Chicken does not Granger cause eggs in all four models.
- But the eggs do Granger cause when the model incorporates at least two lag variables.
- This result is consistent with the conclusion made by Thurman and Fisher (1988).
- But as we see before, the variables may have a nonstationary problem. If it has a unit roots process, all F or Y tests are invalid and the result, therefore, is implausible.